



CREATING A PRIVATE SUBNET

By Ahmed Umar Rehman | September 2026

The screenshot shows the AWS Management Console interface for creating a new subnet. The breadcrumb navigation at the top indicates the path: VPC > Subnets > Create subnet. The page title is 'Subnet settings', with a subtitle 'Specify the CIDR blocks and Availability Zone for the subnet.'

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Input field:
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Dropdown menu:

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
Dropdown menu:

IPv4 subnet CIDR block
Input field: 256 IPs
Navigation buttons: < > ^ v

Tags - optional

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Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a logically isolated virtual network in AWS. It is useful because it allows me to control my resources, networking, and security.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create and configure public and private subnets with routing and network security controls.

One thing I didn't expect in this project was...

One thing I didn't expect in this project is how many separate networking components are needed to properly configure a private subnet.

This project took me...

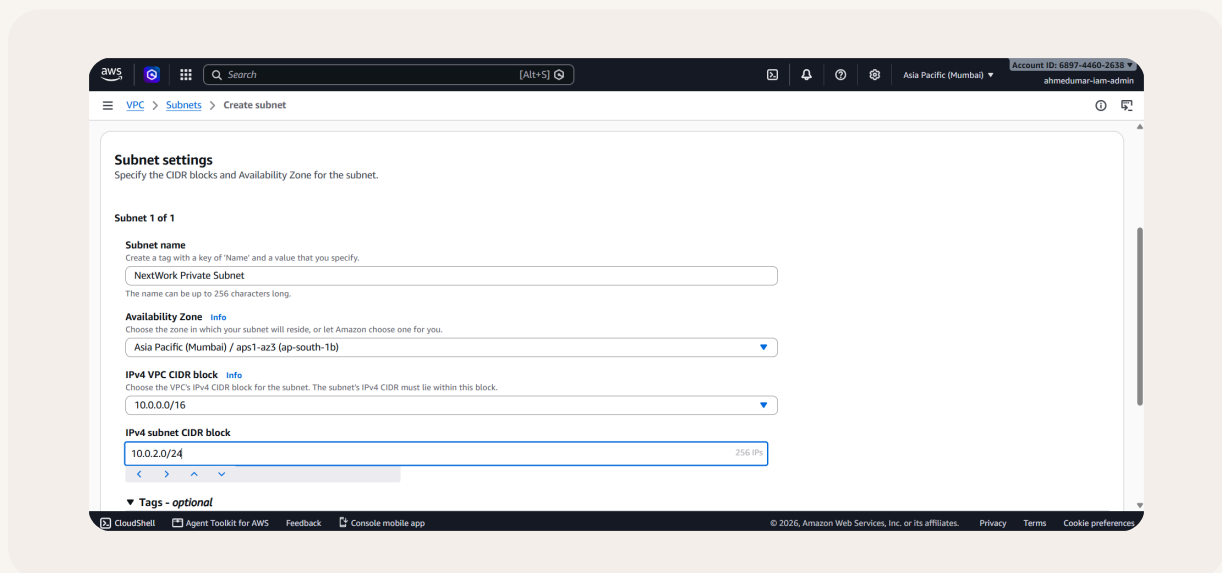
This project took me 30 mins

Private vs Public Subnets

The difference between public and private subnets is that a public subnet has a route to an Internet Gateway, while a private subnet does not have a direct route to the Internet Gateway.

Having private subnets are useful because they keep resources isolated from direct internet access, improving security and reducing exposure to external threats.

My private and public subnets cannot have the same CIDR block because subnet IP ranges within a VPC cannot overlap.

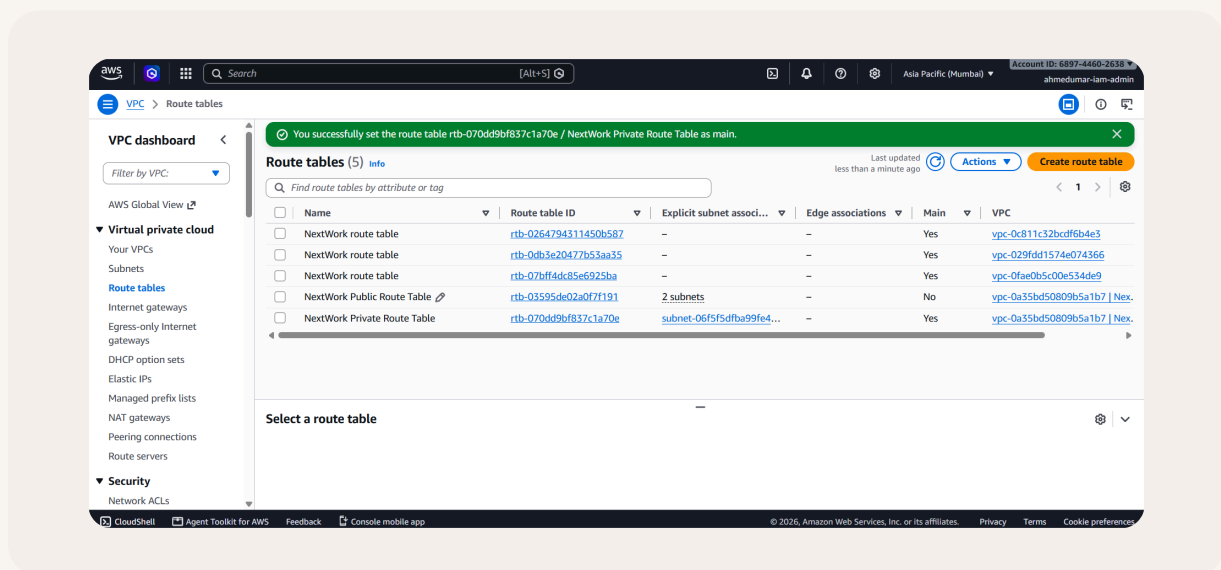


A dedicated route table

By default, my private subnet is associated with 10.0.2.0/24

I had to set up a new route table because the private subnet needs its own routing rules separate from the public subnet.

My private subnet's dedicated route table only has one inbound and one outbound rule that allows local traffic within the VPC.



A new network ACL

By default, my private subnet is associated with the default Network AC 0.0.0.0/0

I set up a dedicated network ACL for my private subnet because it allows me to define specific traffic rules for that subnet.

My new network ACL has two simple rules — one to allow inbound traffic and one to allow outbound traffic.

